

3.1

Honors Geometry · Ch. 3

parallel lines

3.1

Honors Geometry · Ch. 3

perpendicular lines

3.1

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skew lines

3.1

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parallel planes

3.1

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transversal

3.1

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Parallel Postulate

3.1

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**Perpendicular
Postulate**

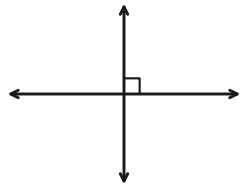
3.2

Honors Geometry · Ch. 3

corresponding angles

perpendicular lines

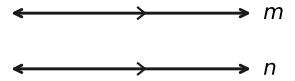
Two lines that intersect to form a right angle.



$$m \perp n$$

parallel lines

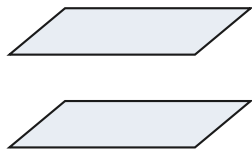
Two lines in the SAME plane that never intersect.



$$m \parallel n$$

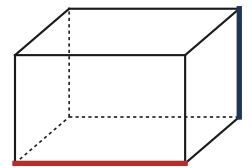
parallel planes

Two planes that never intersect.



skew lines

Lines that do NOT intersect and are NOT coplanar.



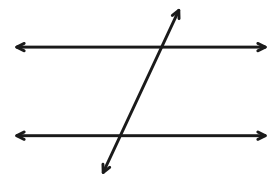
"Not intersecting" alone does not mean parallel.

Parallel Postulate

If there is a line and a point not on the line, then there is exactly ONE line through the point PARALLEL to the given line.

transversal

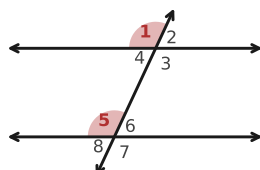
A line that intersects two or more coplanar lines at different points.



Two lines + a transversal make 8 angles.

corresponding angles

Angles in matching corners at the two intersections.



$$\angle 1 \text{ and } \angle 5$$

Perpendicular Postulate

If there is a line and a point not on the line, then there is exactly ONE line through the point PERPENDICULAR to the given line.

3.2

Honors Geometry · Ch. 3

**alternate interior
angles**

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**alternate exterior
angles**

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**consecutive interior
angles**

3.2

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**Corresponding Angles
Postulate**

3.2

Honors Geometry · Ch. 3

**Alternate Interior
Angles Theorem**

3.2

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**Alternate Exterior
Angles Theorem**

3.2

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**Consecutive Interior
Angles Theorem**

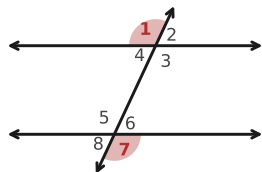
3.3

Honors Geometry · Ch. 3

converse

alternate exterior angles

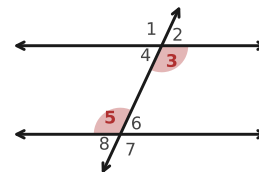
Exterior angles (outside the lines) on OPPOSITE sides of the transversal.



$\angle 1$ and $\angle 7$

alternate interior angles

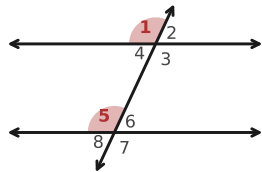
Interior angles (between the lines) on OPPOSITE sides of the transversal.



$\angle 3$ and $\angle 5$

Corresponding Angles Postulate

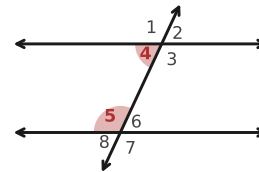
If two parallel lines are cut by a transversal, then the pairs of corresponding angles are CONGRUENT.



$\parallel \rightarrow \angle 1 \cong \angle 5$

consecutive interior angles

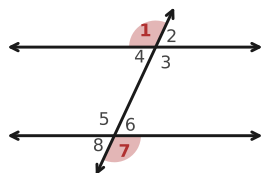
Interior angles on the SAME side of the transversal. Also called same-side interior.



$\angle 4$ and $\angle 5$

Alternate Exterior Angles Theorem

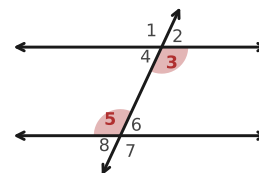
If two parallel lines are cut by a transversal, then the pairs of alternate exterior angles are CONGRUENT.



$\parallel \rightarrow \angle 1 \cong \angle 7$

Alternate Interior Angles Theorem

If two parallel lines are cut by a transversal, then the pairs of alternate interior angles are CONGRUENT.



$\parallel \rightarrow \angle 3 \cong \angle 5$

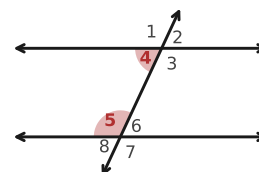
converse

The statement formed by exchanging the hypothesis and the conclusion of a conditional.

If p , then $q \rightarrow$ If q , then p

Consecutive Interior Angles Theorem

If two parallel lines are cut by a transversal, then the pairs of consecutive interior angles are SUPPLEMENTARY.



$\parallel \rightarrow m\angle 4 + m\angle 5 = 180^\circ$

3.3

Honors Geometry · Ch. 3

**Converses: proving
lines parallel**

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flow proof

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paragraph proof

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**Transitive Property of
Parallel Lines**

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Honors Geometry · Ch. 3

**distance from a point
to a line**

3.4

Honors Geometry · Ch. 3

perpendicular bisector

3.4

Honors Geometry · Ch. 3

equidistant

3.4

Honors Geometry · Ch. 3

**Linear Pair
Perpendicular Theorem**

flow proof

A proof that uses boxes and arrows to show the flow of a logical argument.

Each box: a statement. Under it: the reason.

Converses: proving lines parallel

If corresponding, alternate interior, or alternate exterior angles are CONGRUENT — or consecutive interior angles are SUPPLEMENTARY — then the lines are PARALLEL.

Angles given $\rightarrow$ prove $\parallel$ (the 3.2 theorems run backward)

Transitive Property of Parallel Lines

If two lines are parallel to the same line, then they are parallel to each other.

$p \parallel q$ and $q \parallel r \rightarrow p \parallel r$

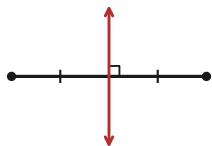
paragraph proof

A proof written as a paragraph of sentences instead of a table.

Same logic as a two-column proof, in sentences.

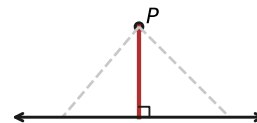
perpendicular bisector

A line perpendicular to a segment at the segment's midpoint.



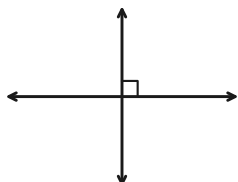
distance from a point to a line

The length of the PERPENDICULAR segment from the point to the line — the shortest distance.



Linear Pair Perpendicular Theorem

If two lines intersect to form a linear pair of CONGRUENT angles, then the lines are PERPENDICULAR.



Each angle is 90° .

equidistant

The same distance from two or more objects.

Every point on a perpendicular bisector is equidistant from the endpoints.

3.4

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Perpendicular Transversal Theorem

3.4

Honors Geometry · Ch. 3

Lines Perpendicular to a Transversal Theorem

3.5

Honors Geometry · Ch. 3

slope

3.5

Honors Geometry · Ch. 3

parallel lines (slopes)

3.5

Honors Geometry · Ch. 3

perpendicular lines (slopes)

3.5

Honors Geometry · Ch. 3

coinciding lines

3.5

Honors Geometry · Ch. 3

point-slope form

3.5

Honors Geometry · Ch. 3

slope-intercept form

Lines Perpendicular to a Transversal Theorem

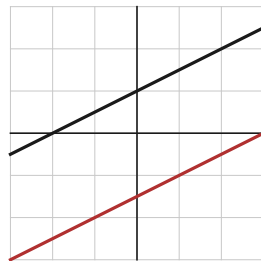
In a plane, if two lines are perpendicular to the SAME line, then they are PARALLEL to each other.

Perpendicular Transversal Theorem

In a plane, if a transversal is perpendicular to one of two PARALLEL lines, then it is perpendicular to the OTHER.

parallel lines (slopes)

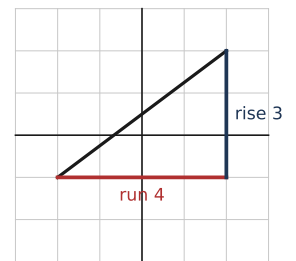
Same slope, different y-intercepts.



$$y = \frac{1}{2}x + 1 \text{ and } y = \frac{1}{2}x - 3$$

slope

The ratio of vertical change to horizontal change. $m = (y_2 - y_1) \div (x_2 - x_1)$.



Horizontal: $m = 0$. Vertical: undefined.

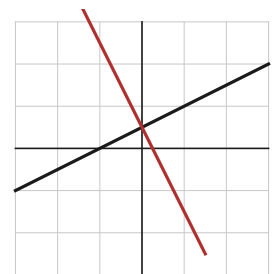
coinciding lines

Same slope AND same y-intercept — they are the same line written two ways.

$$y = 2x + 4 \text{ and } 2x - y = -4$$

perpendicular lines (slopes)

Opposite reciprocal slopes — the product of the slopes is -1 .



$$m = \frac{1}{2} \text{ and } m = -2$$

slope-intercept form

$y = mx + b$ — m is the slope, b is the y-intercept.

Vertical line: $x = a$. Horizontal line: $y = b$.

point-slope form

$y - y_1 = m(x - x_1)$ — uses a point (x_1, y_1) and the slope m .

Through $(2, -1)$, $m = 3$: $y + 1 = 3(x - 2)$